Cyclohexane Reactor Ver 1 · operator training simulator

Plot plan, elevations and piping isometrics

Document no. CHX-V1-PL-PP-001 · **Discipline** Layout / piping

Client Department of Chemical Engineering, Faculty of Engineering, Chulalongkorn University (teaching)

- Plot plan of the unit and the site: 21 unit items from CHX-V1-ME-EQL-001 placed at scale, 5 off-plot items (tanks, flare); coordinates and levels in section 4 (x east, y north, z up, m; grade EL 0.0).
- Elevations set by hydraulics: P-101 NPSHa 5.57 m from R-101 at EL +7.0; gravity reflux 3.47 m of head for 2.75 m needed; condenser deck EL +18.5.
- Five isometric sketches (benzene feed, pump-around, overhead / reflux / vent, recycle hydrogen, relief and flare) with bills of materials; 52 pipe runs, 2 384 m in total.
- layout.json for the Visual plant 3D tab: 98 valves (49 hand valves), 40 field instruments, 13 platforms and an outside-operator round of 14 stations.
- Checks: 92 rows, 89 ✓, 3 ✗ — each ✗ carries an engineering query (EQ-la-1 … EQ-la-9).
- Every placement and spacing rule is typical practice for teaching, labelled assumed; no vendor, site or contractor drawing is implied.

Revisions

Rev	Date	Description	Originator	Checker	Approver
0	2026-10-26	Issued for teaching	Apinan Soottitantawat		

This document is generated for teaching from the simulator model of the unit. It is not issued for construction; every number comes from the model or is labelled assumed / typical practice. No client, contractor or vendor is implied.

Contents

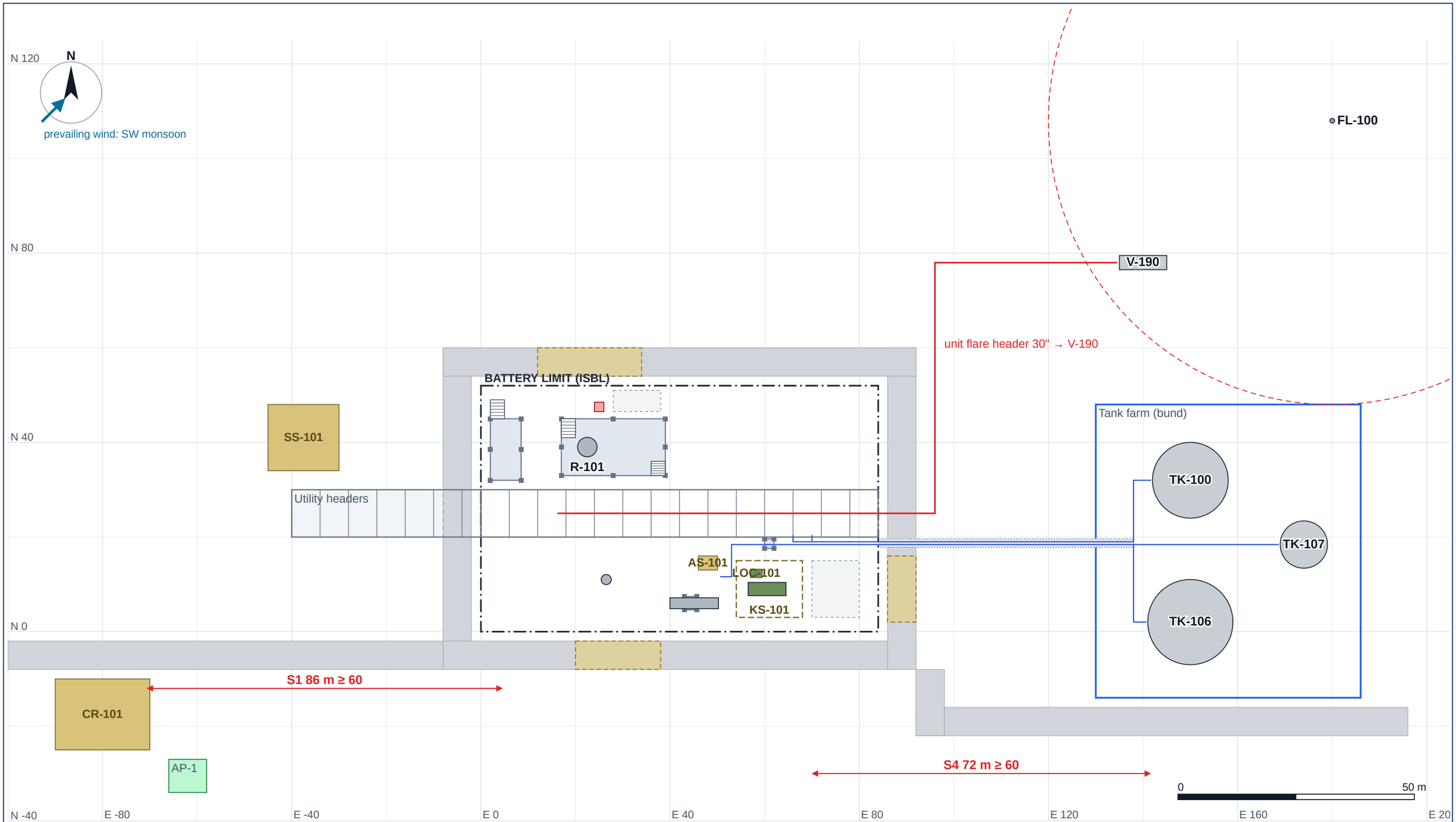
1 · Basis and placement rules 3
2 · Site plan (sheet DWG-1) 4
3 · Unit plot plan (sheet DWG-2) 5
3.1 · Hazardous-area overlay and pipe routes (sheet DWG-3) 6
4 · Coordinates and levels 7
5 · Elevations and levels (sheet DWG-4) 8
5.1 · Section B–B (sheet DWG-5) 9
5.2 · Elevations from hydraulics 10
5.3 · Platforms and access 10
5.4 · Where each field instrument is read and each hand valve operated 10
6 · Isometrics — ISO-01 Benzene feed: P-100A/B → FV-101 / XV-101 → R-101 N1 13
ISO-01 · bill of materials (per line, from the route) 14
6.2 · ISO-02 Pump-around loop: R-101 → P-101A/B → FV-104 → E-101 / by-pass → R-101 15
ISO-02 · bill of materials (per line, from the route) 16
6.3 · ISO-03 Overhead, reflux and vent: R-101 → E-103 → V-103 → reflux to R-101, gas to R-102 17
ISO-03 · bill of materials (per line, from the route) 18
6.4 · ISO-04 Recycle hydrogen: V-104 → K-101 → rack → FIC-103 / FV-103 → R-101 sparger 19
ISO-04 · bill of materials (per line, from the route) 20
6.5 · ISO-05 Relief and flare: PSV tailpipes, EDV-101 / HV-121 and the unit flare header to V-190 21
ISO-05 · bill of materials (per line, from the route) 22
7 · layout.json and the outside operator's round (sheet DWG-6) 23
7.1 · Route stations 24
7.2 · layout.json keys 24
8 · Checks against the other documents 25
Every EQL item on the plot plan 25
Every line number on an isometric is in the line list or the EPC P&ID 25
Spacing rules (typical practice, assumed) 26
Hazardous-area zones (CHX-V1-EE-HAC-001) on the plan 26
layout.json validation (also tools/validate_layout.js) 26
Elevations from hydraulics 26
Cross-document 26
9 · Engineering queries and assumptions 28
Assumptions 28

1 · Basis and placement rules

The plot plan places the items of the equipment list CHX-V1-ME-EQL-001 with their datasheet sizes, the lines of the line list CHX-V1-PE-LL-001 and the EPC P&ID CHX-V1-PID-001, the valves and field instruments of the EPC P&ID, the zones of the hazardous-area schedule CHX-V1-EE-HAC-001 and the relief devices of CHX-V1-PE-RLB-001. Placement follows typical practice (assumed, below). Prevailing wind: SW monsoon (May–October), prevailing (assumed typical for central Thailand); the control room is upwind of it.

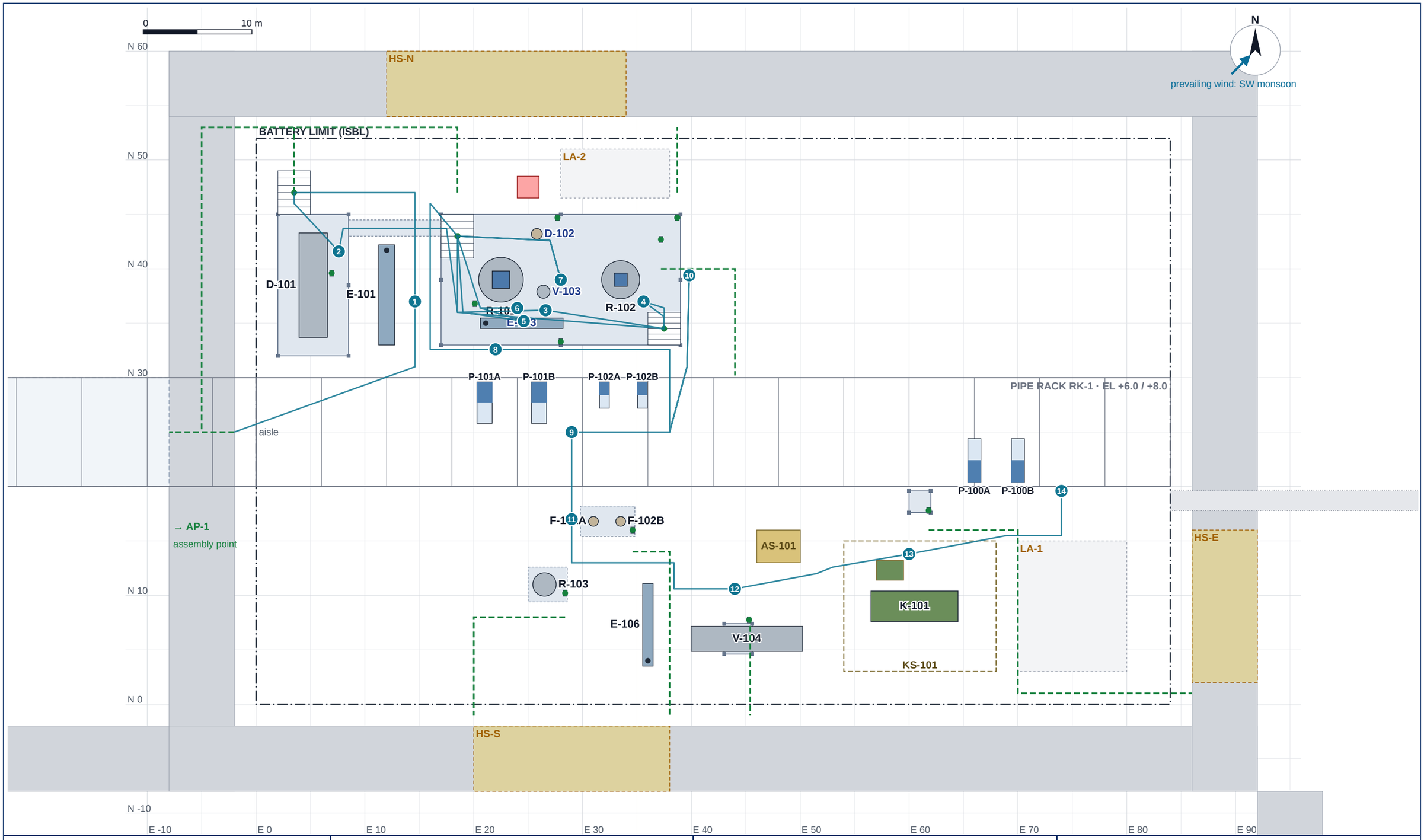
Rule (typical practice, assumed)	Minimum / limit	This plot plan	Item	
S1 Occupied control room (not blast-rated) to process equipment	≥ 60	85.9	D-101	✓
S2 Control room to atmospheric storage tanks	≥ 60	211.3	TK-106	✓
S3 Substation / LER to process equipment (outside every zone)	≥ 30	34.0	D-101	✓
S4 Process equipment to atmospheric flammable storage tanks	≥ 60	71.8	P-100B – TK-100	✓
S5 Tank to tank, shell to shell (fixed roof, ≥ ½ the larger diameter)	≥ D/2	4.0	TK-100 – TK-106: 13.0 m vs 9.0 m	✓
S6 Flare stack to process equipment and tanks (sterile radius, radiation study to confirm)	≥ 60	73.7	TK-100	✓
S7 Hydrogen compressor to the reactors (large hot hydrocarbon inventories)	≥ 15	28.9	K-101 – R-103	✓
S8 V-104 close to K-101 (short suction line, LSHH-104 protection)	≤ 15	6.3	V-104 – K-101	✓
S9 Pumps handling flammables to the vessels they serve	≥ 3	7.4	P-101A – R-101	✓
S10 Clear access between adjacent items (maintenance)	≥ 1.5	1.6	F-102A – F-102B	✓
S11 Process equipment to the battery limit (roads outside)	≥ 2.5	3.5	E-106	✓
S12 Tube bundle pulling / rodding space in front of each channel (tube length + 1 m; roads count)	≥ L+1	21.7	E-103 30.0 ≥ 7.1 m · E-101 30.0 ≥ 8.3 m · E-106 30.0 ≥ 7.1 m	✓
S13 Two escape routes from every platform; travel to a stair or ladder ≤ 25 m	≤ 25	13.0	11 walkable areas: 5 with ≥ 2 escapes, 6 small with one ladder	✓
S14 Control room upwind: within ±45° of the prevailing-wind source (SW monsoon, from 225°)	≤ 45	25.4	CR-101 bearing 250° from the unit centre; SW source 225°	✓
S15 Atmospheric vent PSV-109: ≥ 3 m above any platform within 15 m	≥ 3	3.0	vent top EL +21.5; highest platform within 15 m EL +18.5 (PF-D80, PF-D112, PF-BR, PF-R45, PF-R80, PF-R110, PF-R140, PF-E185)	✓
S16 Every PSV reachable from a platform or grade (maintenance)	every PSV	5	PSV-101 (PF-R140), PSV-109 (PF-D112), PSV-102 (PF-R2T), PSV-104 (PF-V104), PSV-112 (PF-K112)	✓
S17 Clear height under the pipe rack lower tier (vehicle and crane passage)	≥ 5	5.5	lower tier EL +6 less 0.5 m beam	✓
S18 Analyser shelter within 20 m of its sample points (sample transport lag)	≤ 20	1.5	AS-101 to 10"-G-2007 / 6"-P-2011	✓

2 · Site plan (sheet DWG-1)



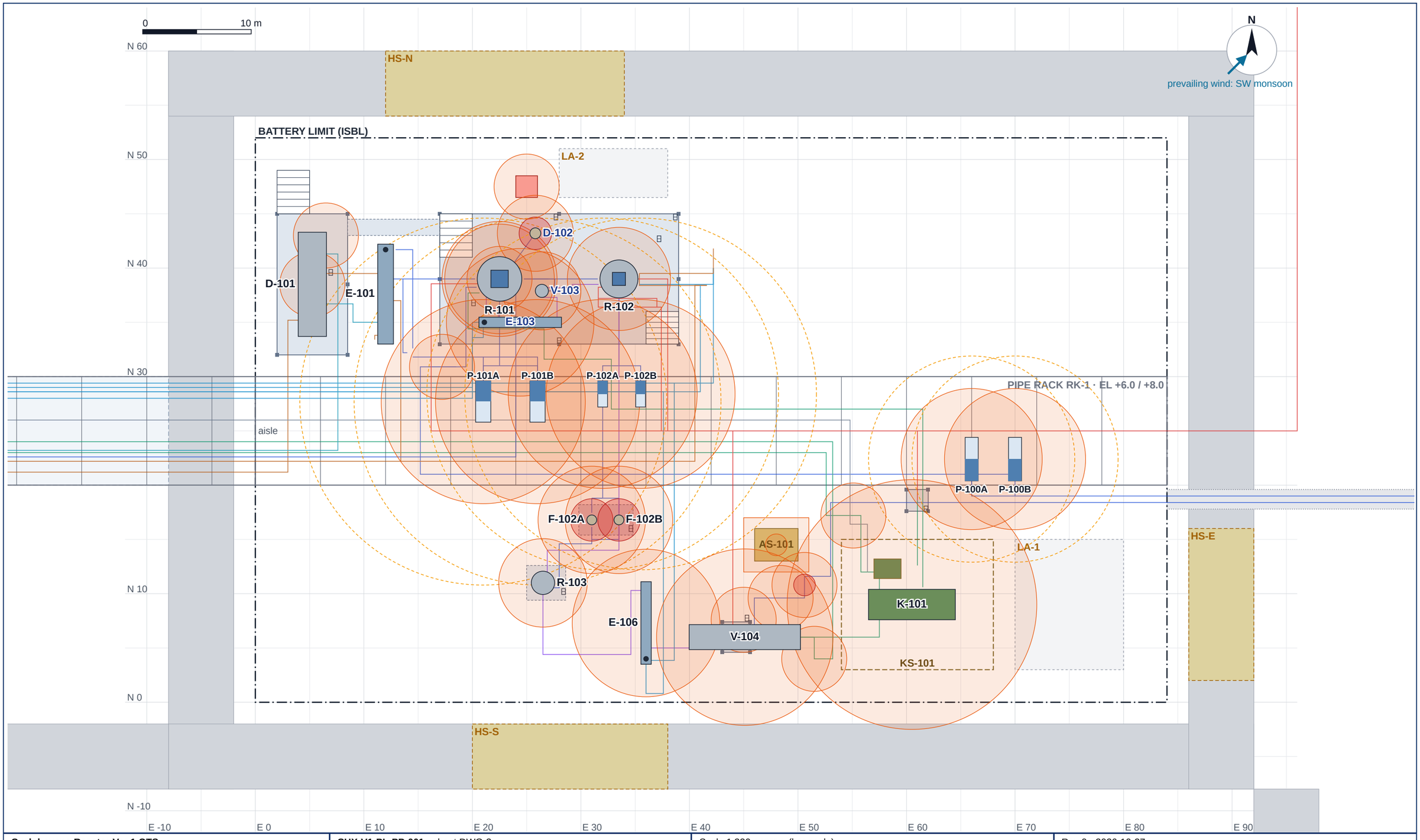
Cyclohexane Reactor Ver 1 OTS Issued for teaching · numbers from the simulator model	**CHX-V1-PL-PP-001** · sheet DWG-1 Site plan: unit, utilities, control room, tank farm and flare (spacing dimensions)	Scale 1:780 approx. (bar scale) Units m · EL above grade 0.0 · plant north = true north (assumed)	Rev 0 · 2026-10-27 Drawn Apinan Soottitantawat · checked — · approved —

3 · Unit plot plan (sheet DWG-2)



Cyclohexane Reactor Ver 1 OTS Issued for teaching · numbers from the simulator model	**CHX-V1-PL-PP-001** · sheet DWG-2 Unit plot plan with escape routes and the operator round	Scale 1:330 approx. (bar scale) Units m · EL above grade 0.0 · plant north = true north (assumed)	Rev 0 · 2026-10-27 Drawn Apinan Soottitantawat · checked — · approved —

3.1 · Hazardous-area overlay and pipe routes (sheet DWG-3)



Cyclohexane Reactor Ver 1 OTS Issued for teaching · numbers from the simulator model	**CHX-V1-PL-PP-001** · sheet DWG-3 Unit plot plan with the hazardous-area overlay (CHX-V1-EE-HAC-001) and pipe routes	Scale 1:330 approx. (bar scale) Units m · EL above grade 0.0 · plant north = true north (assumed)	Rev 0 · 2026-10-27 Drawn Apinan Soottitantawat · checked — · approved —

4 · Coordinates and levels

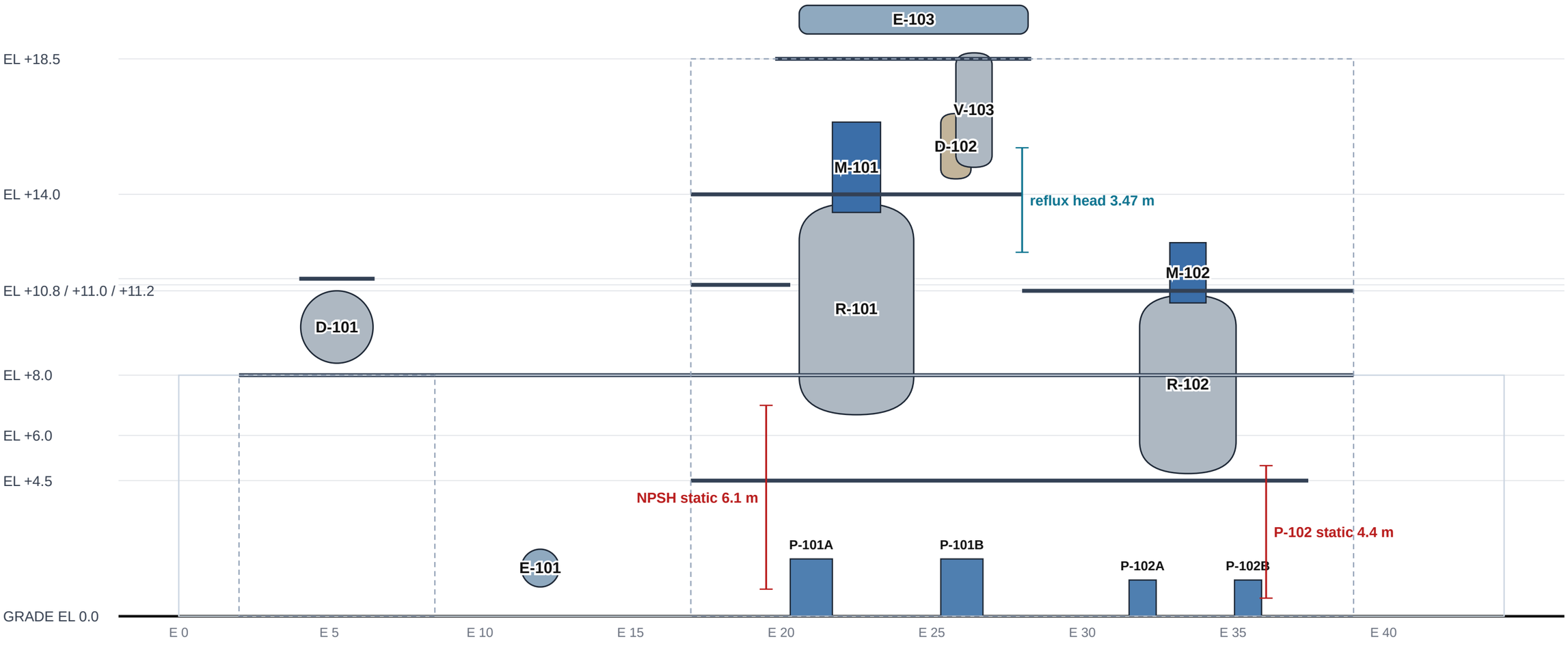
Centre of each item; bottom and top elevations of its envelope (insulation included). Buildings:

Tag	Building	x	y	height
CR-101	Central control room (occupied)	-90 … -70	-25 … -10	5.0
SS-101	Substation, MCC and local equipment room (DCS/SIS marshalling)	-45 … -30	34 … 48	6.0
KS-101	K-101 compressor shelter (open sides, roof EL +10.0, 15 t overhead crane on rails at EL +8.5)	54 … 68	3 … 15	10.0
AS-101	Analyser shelter (AT-100, AT-103, AT-105; ventilated, gas-detected)	46 … 50	13 … 16	3.0
LOC-101	K-101 lube-oil and seal-gas console (package auxiliary)	57 … 59.5	11.4 … 13.2	2.2

Tag	Item	Type	x	y	bottom EL	top EL	Plan size m	Area	Support / basis
R-101	Main hydrogenation reactor	vcyl	22.50	39.00	7.00	13.40	4.10 × 4.10	Reactor structure	four lugs on the EL +8.0 floor beams of the reactor structure · NPSH of P-101 (bubble-point liquid): lowest point EL +7.0
M-101	R-101 agitator drive (motor, gearbox)	box	22.50	39.00	13.40	16.40	1.60 × 1.60	Reactor structure	on the R-101 top flange
R-102	Finishing slurry reactor (jacketed)	vcyl	33.50	39.00	5.00	10.40	3.50 × 3.50	Reactor structure	four lugs on the EL +8.0 floor beams (common structure with R-101) · gravity overflow from R-101 (N7 above N1) and NPSH of P-102
M-102	R-102 agitator drive (HOLD)	box	33.50	39.00	10.40	12.40	1.20 × 1.20	Reactor structure	on the R-102 top flange
V-103	Reflux drum	vcyl	26.40	37.90	15.00	18.60	1.20 × 1.20	Reactor structure	skirt 1.0 m on the EL +14.0 deck · gravity reflux to R-101 N4 against the overhead pressure drop
E-103	Reflux condenser	exchanger	24.40	35.00	19.32	20.27	7.60 × 0.95	Reactor structure	saddles 0.8 m on the EL +18.5 condenser deck · shell outlet above the V-103 inlet: free-draining condensate
D-102	Catalyst charge pot	vcyl	25.80	43.20	14.60	16.60	1.00 × 1.00	Reactor structure	legs on the EL +14.0 deck under the catalyst hoist · charge to R-101 by gravity and N_2 pressure; drums lifted by the monorail
E-101	Pump-around cooler / LP steam generator	exchanger	12.00	37.60	0.88	2.33	1.45 × 9.20	Steam bay	two saddles on concrete piers at grade · below D-101 for natural circulation; channel to the north for tube cleaning
D-101	Steam drum	hcyl	5.25	38.50	8.30	10.90	2.60 × 9.60	Steam bay	saddles on the EL +8.0 deck of the steam-drum structure · normal water level ≈ 7.9 m above the E-101 centre line (thermosyphon head)
R-103	Trickle-bed finishing reactor	column	26.50	11.00	2.00	10.38	2.15 × 2.15	South (polishing and separation)	skirt on a concrete foundation · outlet free-draining to E-106; catalyst loading from the top by crane (south road)
F-102A	Catalyst candle filter A (duty)	vcyl	31.00	16.80	0.60	3.40	0.90 × 0.90	South (polishing and separation)	legs on a paved, kerbed area draining to the closed drain
F-102B	Catalyst candle filter B (standby)	vcyl	33.50	16.80	0.60	3.40	0.90 × 0.90	South (polishing and separation)	legs on a paved, kerbed area draining to the closed drain
E-106	Product cooler	exchanger	36.00	7.30	0.93	1.88	0.95 × 7.60	South (polishing and separation)	two saddles on concrete piers · channel toward the road for bundle pulling
V-104	HP separator	hcyl	45.10	6.00	1.45	3.75	10.25 × 2.30	South (polishing and separation)	two saddles on concrete piers · close to K-101 suction; level bridle readable from grade
K-101	Recycle hydrogen compressor package (compressor, gear, motor)	compressor	60.50	9.00	0.00	3.60	8.00 × 2.80	Compressor shelter	block foundation on its own piles (vibration), inside the open-sided shelter KS-101 with an overhead crane · near V-104 (short suction line); crane and laydown for maintenance
P-100A	Benzene feed pump A	pump	66.00	22.40	0.00	1.50	1.20 × 4.00	Pump row under the pipe rack	baseplate on a concrete plinth, kerbed and drained · south row: suction from the pipe track (south), motor toward the rack aisle
P-100B	Benzene feed pump B (spare)	pump	70.00	22.40	0.00	1.50	1.20 × 4.00	Pump row under the pipe rack	baseplate on a concrete plinth, kerbed and drained · south row: suction from the pipe track (south), motor toward the rack aisle
P-101A	Pump-around pump A	pump	21.00	27.70	0.00	1.90	1.40 × 3.80	Pump row under the pipe rack	baseplate on a concrete plinth, kerbed and drained · north row: suction from the north (reactor side), motor toward the rack aisle
P-101B	Pump-around pump B (spare)	pump	26.00	27.70	0.00	1.90	1.40 × 3.80	Pump row under the pipe rack	baseplate on a concrete plinth, kerbed and drained · north row: suction from the north (reactor side), motor toward the rack aisle
P-102A	R-102 transfer pump A	pump	32.00	28.40	0.00	1.20	0.90 × 2.40	Pump row under the pipe rack	baseplate on a concrete plinth, kerbed and drained · north row: suction from the north (reactor side), motor toward the rack aisle
P-102B	R-102 transfer pump B (spare)	pump	35.50	28.40	0.00	1.20	0.90 × 2.40	Pump row under the pipe rack	baseplate on a concrete plinth, kerbed and drained · north row: suction from the north (reactor side), motor toward the rack aisle
TK-100	Benzene feed tank (fixed roof, internal floater)	tank	150.00	32.00	0.30	12.30	16.00 × 16.00	Tank farm (outside battery limit)	ring-wall foundation in the common bund · 48 h of 34.8 t/h at 857.8 kg/m³ = 1947 m³ ≤ 85 % of 2413 m³ (assumed size)
TK-106	Cyclohexane product tank (on spec)	tank	150.00	2.00	0.30	12.30	18.00 × 18.00	Tank farm (outside battery limit)	ring-wall foundation in the common bund · 48 h of 37.5 t/h at 743 kg/m³ = 2422 m³ ≤ 85 % of 3054 m³ (assumed size)
TK-107	Slop / off-spec tank	tank	174.00	18.40	0.30	10.30	10.00 × 10.00	Tank farm (outside battery limit)	ring-wall foundation in the common bund · off-spec and start-up slop (assumed size)
V-190	Flare knock-out drum (outside battery limit)	hcyl	140.00	78.00	0.50	3.50	10.00 × 3.00	Flare system (outside battery limit)	saddles · size not issued (assumed for the plan)
FL-100	Flare stack (outside battery limit)	vcyl	180.00	108.00	0.00	45.00	1.00 × 1.00	Flare system (outside battery limit)	guyed stack · height and sterile radius from a radiation study (not issued; 60 m assumed)

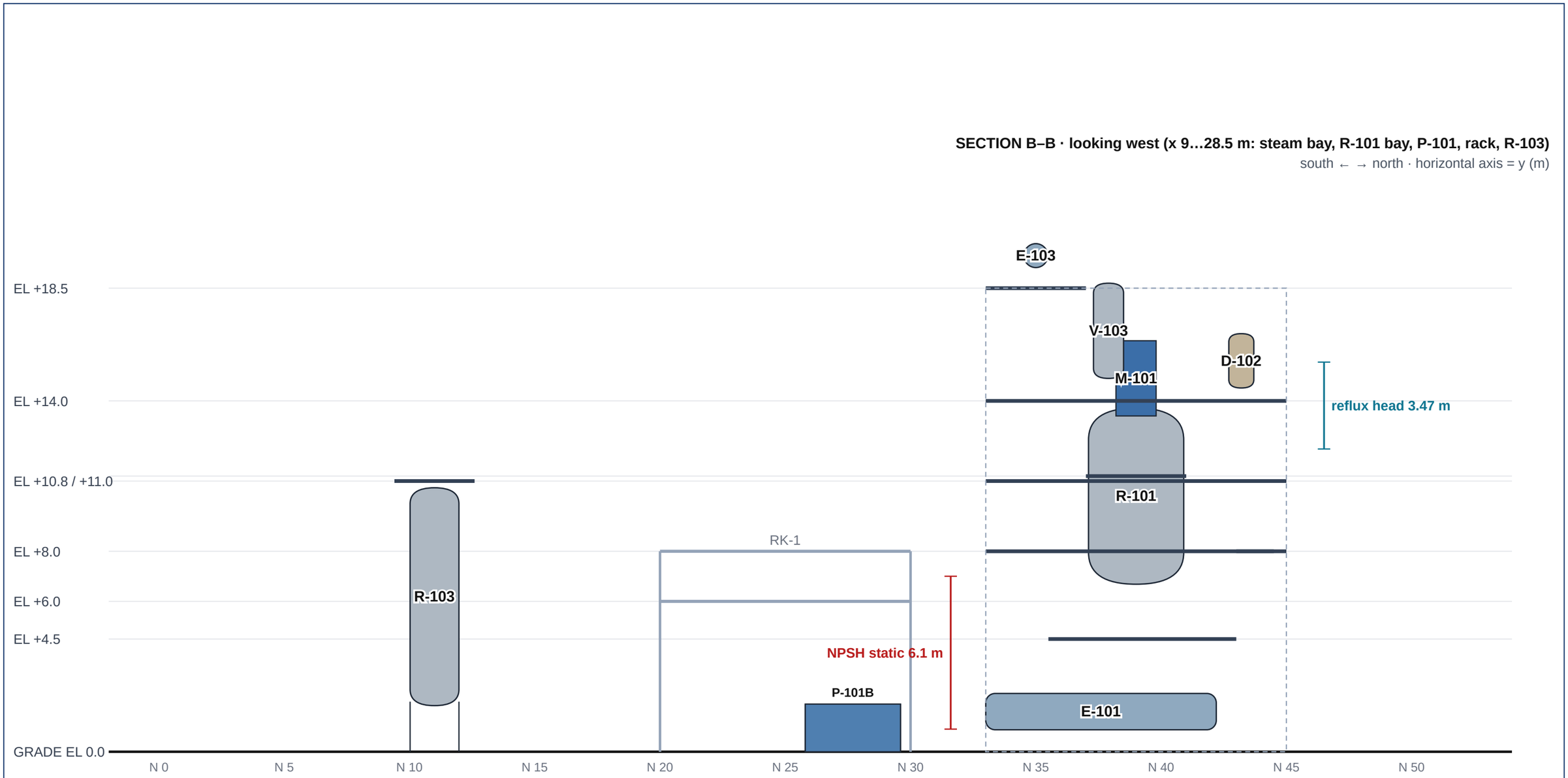
5 · Elevations and levels (sheet DWG-4)

ELEVATION A–A · looking north (items north of the rack and the pump row)
west ← → east · horizontal axis = x (m)



Cyclohexane Reactor Ver 1 OTS Issued for teaching · numbers from the simulator model	**CHX-V1-PL-PP-001** · sheet DWG-4 Elevation A–A looking north (reactor structure, steam bay, pump row)	Scale not to a fixed scale (levels as marked) Units m · EL above grade 0.0 · plant north = true north (assumed)	Rev 0 · 2026-10-27 Drawn Apinan Soottitantawat · checked — · approved —

5.1 · Section B–B (sheet DWG-5)



Cyclohexane Reactor Ver 1 OTS Issued for teaching · numbers from the simulator model	**CHX-V1-PL-PP-001** · sheet DWG-5 Section B–B looking west (steam bay, R-101 bay, P-101, rack, R-103)	Scale not to a fixed scale (levels as marked) Units m · EL above grade 0.0 · plant north = true north (assumed)	Rev 0 · 2026-10-27 Drawn Apinan Soottitantawat · checked — · approved —

5.2 · Elevations from hydraulics

Pumps	From	Vessel low EL	Pump CL EL	Static m	Route / Le m	Losses m	NPSHa m	Needed m	Basis	
P-101A/B	R-101	7.00	0.90	6.10	16.8 / 78.5	0.53	5.57	5.20	R-101 liquid is at its bubble point (dissolved H_2): NPSHa = static head − suction losses; R-101 bottom at EL +7.0 (layout)	✓
P-102A/B	R-102	5.00	0.60	4.40	15.1 / 28.8	0.49	3.91	3.00	R-102 liquid at its bubble point; R-102 bottom at EL +5.0 (layout)	✓
P-100A/B	TK-100	0.30	0.80	-0.50	91.5 / 115.7	0.37	7.99	4.55	atmospheric tank, benzene at 40 °C; vapour pressure from the model correlation PsatC (as the instrument list does for FV-101)	✓

NPSHa ≥ max(NPSHr + 1.0 m, 1.3 × NPSHr) (typical practice); NPSHr values are assumed until the pump datasheets are confirmed

Gravity reflux. V-103 sits downstream of R-101 in the gas path, so it is at a LOWER pressure than R-101: its liquid must stand high enough above N4 to push the reflux back against the overhead pressure drop. ΔP R-101 → V-103 = lines 10"-PG-2001-B2 0.0291 + 8"-PG-2002-B2 0.0251 + E-103 shell 0.1 bar (assumed) = 0.154 bar = 2.14 m of reflux (ρ 735.3 kg/m³); reflux line and FT-105 0.12 m; margin 0.5 m → needed 2.75 m; V-103 low level EL +15.55 − R-101 N4 EL +12.08 = **3.47 m** ✓.
Overflow R-101 → R-102. 4"-P-1008-B3 falls 0.85 m over 6.8 m (12.5 %); the flattest leg 10.2 % ≥ 1 % (self-draining slurry, typical). **Thermosyphon.** D-101 normal water level EL +9.48, E-101 CL EL +1.60. **PSV-109 vent** top EL +21.5, 3.0 m above the highest platform within 15 m.

5.3 · Platforms and access

Platform	EL	Extent x / y m	Escape down by	Note
PF-D80 · D-101 deck EL +8.0	8.0	2…8.5 / 32…45	ST-1, ST-2, LD-1	
PF-D112 · D-101 top platform EL +11.2 (PSV-109, PT-110 nozzle)	11.2	4…6.5 / 37…40	LD-5	small platform, one ladder (typical)
PF-BR · Bridge EL +8.0 (steam bay ↔ reactor structure)	8.0	8.5…17 / 43…44.5	ST-1, ST-2, LD-1	
PF-R45 · Reactor bottoms platform EL +4.5 (bottom nozzles, EIV-101, EIV-102, drains)	4.5	17…37.5 / 35.5…43	ST-1, LD-4	
PF-R80 · Reactor floor EL +8.0 (vessel supports, side nozzles, LV-101 station, HV-113)	8.0	17…39 / 33…45	ST-1, ST-2, LD-1	
PF-R110 · R-101 manway platform EL +11.0	11.0	17…20.3 / 37…41	ST-1	small platform, one ladder (typical)
PF-R2T · R-102 top platform EL +10.8 (agitator, PSV-102, EDV-101, HV-121)	10.8	28…39 / 33…45	ST-3, LD-1	
PF-R140 · Reactor top deck EL +14.0 (M-101, PSV-101, D-102, V-103, catalyst hoist)	14.0	17…28 / 33…45	ST-1, LD-2	
PF-E185 · Condenser deck EL +18.5 (E-103)	18.5	19.8…28.3 / 33…37	LD-3, LD-3B	
PF-R3T · R-103 top platform EL +10.8 (inlet distributor, manway, catalyst loading)	10.8	25…28.6 / 9.4…12.6	LD-6	small platform, one ladder (typical)
PF-F2 · Filter platform EL +2.2 (F-102 covers, PDT-102)	2.2	29.8…34.8 / 15.4…18.2	LD-9	small platform, one ladder (typical)
PF-V104 · PSV-104 platform EL +8.7	8.7	43…45.6 / 4.6…7.4	LD-7	small platform, one ladder (typical)
PF-K112 · PSV-112 platform EL +7.6	7.6	60…62 / 17.6…19.6	LD-8	small platform, one ladder (typical)

5.4 · Where each field instrument is read and each hand valve operated

Tag	Service	x	y	EL	From	Normal	Station
FT-101	Benzene feed to R-101 → FV-101 (fail closed)	15.6	30.9	1.0	grade		8
TT-100	R-101 temperature, independent element (SIS element)	20.2	37.9	9.0	PF-R80		3
TT-101	R-101 temperature → cascade to FIC-104 pump-around through E-101	20.4	40.2	9.0	PF-R80		3
PT-101	R-101 pressure (SIS element)	22.1	37.8	13.6	PF-R140		5
TT-103	E-103 reflux condenser outlet / V-103 (reflux and vent gas temperature)	26.6	38.2	19.6	PF-E185		6
LG-103	V-103 reflux drum level, local	27.1	37.9	16.3	PF-R140		5
FT-105	Reflux from V-103 back to R-101 (the product R-101 boils off and condenses)	27.1	37.9	14.6	PF-R140		5
JT-101	Agitator M-101 power (design ≈150 kW gassed)	-37.5	41.0	1.5	SS-101		–
SSL-101	Agitator shaft speed (diverse from the M-101 run contact)	23.4	39.0	14.6	PF-R140		5
LT-101	R-101 level → LV-101 overflow to R-102 (fail closed)	22.5	40.2	13.6	PF-R140		5
FT-103	Hydrogen (recycle + make-up) to R-101 sparger → FV-103 (fail closed)	32.3	31.6	1.0	grade		8
FT-104	Pump-around flow through E-101 → FV-104 (fail open)	14.5	36.0	1.2	grade		1
TT-104	Pump-around return from E-101 to R-101	12.0	41.9	3.0	grade		1
FT-111	LP steam exported from D-101 to the 3.5 bar g header	3.0	33.0	8.9	PF-D80		2
PT-110	D-101 steam pressure → PV-110 LP steam export (fail open)	6.6	36.6	9.4	PF-D80		2
FT-109	Boiler feed water to D-101	7.6	33.2	8.9	PF-D80		2
LT-109	D-101 steam drum level → LV-109 boiler feed water (fail open)	6.8	39.5	9.2	PF-D80		2
FT-108	MP steam to the R-102 jacket (HEAT mode)	40.5	35.4	1.2	grade		10
PT-102	R-102 pressure	33.5	38.0	11.2	PF-R2T		4
TT-102	R-102 temperature → TV-102 jacket (MP steam in HEAT, CW in COOL; fail closed)	35.4	38.4	8.7	PF-R80		3
LT-102	R-102 level → LV-102 to R-103 through filter F-102 (fail closed)	33.5	40.2	11.0	PF-R2T		4
TT-106	R-103 inlet (R-102 liquid + vapour + hydrogen)	25.0	9.8	1.4	grade		11
PDT-102	Catalyst filter F-102 differential pressure	32.2	15.8	2.9	PF-F2		11
FT-102	R-102 liquid to R-103 through F-102	30.8	14.6	1.0	grade		11
TT-107A	R-103 bed thermocouple, top (cell 1)	25.0	10.1	1.4	grade		11

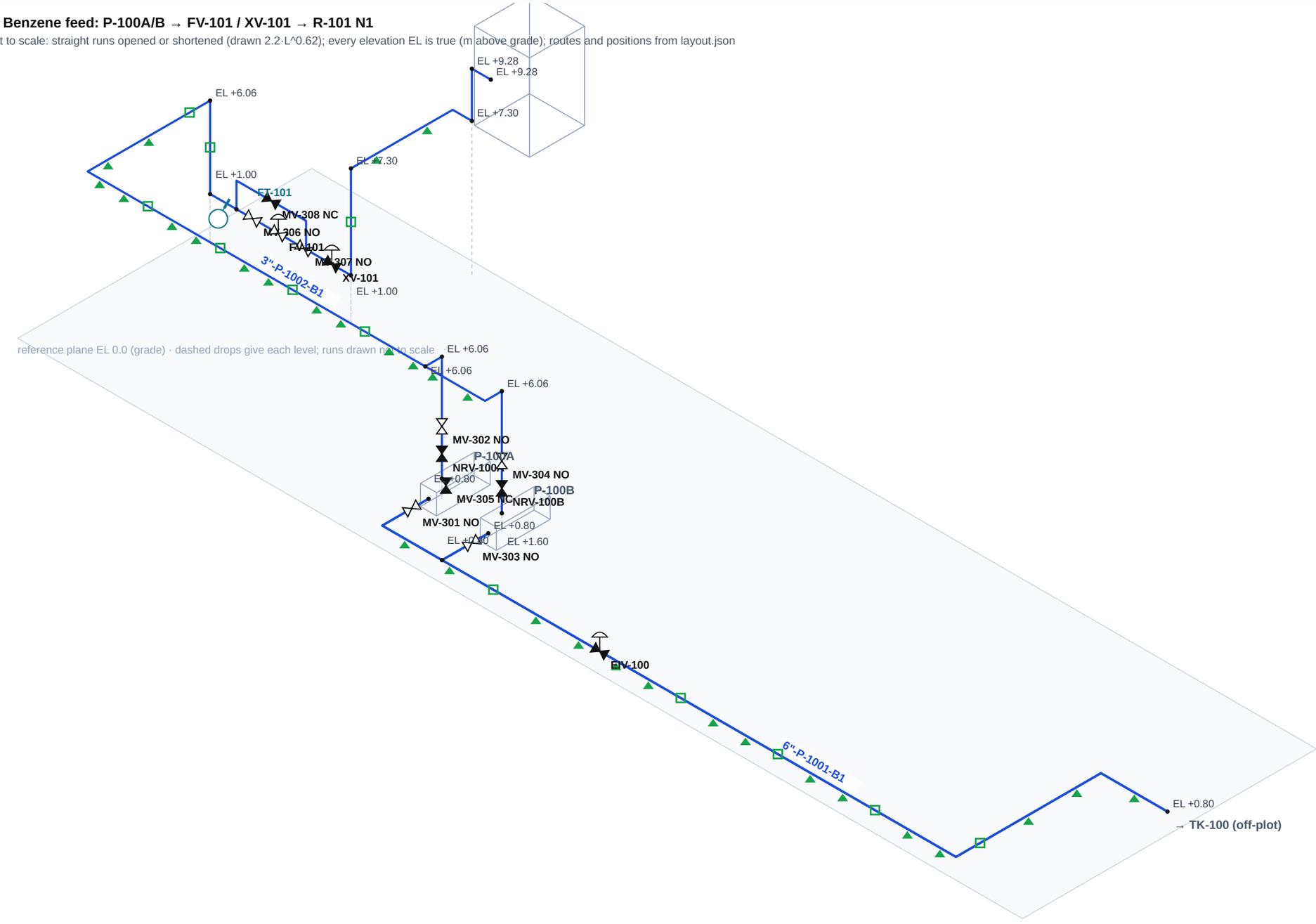
Tag	Service	x	y	EL	From	Normal	Station
TT-107B	R-103 bed thermocouple, cell 2	25.0	10.4	1.4	grade		11
TT-107C	R-103 bed thermocouple, middle (cell 3)	25.0	10.7	1.4	grade		11
TT-107D	R-103 bed thermocouple, cell 4	25.0	11.0	1.4	grade		11
TT-107E	R-103 bed thermocouple, bottom (cell 5 = outlet)	25.0	11.3	1.4	grade		11
FT-106	Make-up hydrogen through PV-105	54.2	17.2	1.0	grade		13
PT-105	Loop pressure at V-104 → make-up hydrogen PV-105 (fail closed)	47.0	7.7	1.4	grade		12
FT-107	Purge from V-104 to fuel gas (hand controller, fail closed)	52.2	4.0	1.2	grade		12
PT-112	K-101 discharge (hydrogen header)	61.5	11.6	1.8	grade		13
AT-100	Loop oxygen (start-up purge analyzer; < 1 % before hydrogen)	46.6	15.4	1.5	AS-101		12
AT-105	Recycle gas hydrogen purity (rest CH4 / N2)	47.6	15.4	1.5	AS-101		12
LT-104	V-104 separator level → LV-104 product to storage (fail closed)	44.0	7.5	1.6	grade		12
LT-104S	V-104 level, SIS element of LSHH-104	46.0	7.5	1.6	grade		12
TT-109	V-104 separator temperature (E-106 outlet)	42.0	7.4	2.0	grade		12
FT-110	Cyclohexane product to storage (HS-115 selects TK-106 on-spec or TK-107 slop)	52.0	11.6	0.9	grade		12
AT-103	Product benzene, on-line GC (4-min cycle, 2-min sample delay; spec ≤ 10 ppm)	48.8	15.4	1.5	AS-101		12
MV-301	P-100A suction	66.0	20.0	0.8	grade	NO	14
MV-302	P-100A discharge	66.0	21.4	2.9	grade	NO	14
MV-303	P-100B suction	70.0	20.0	0.8	grade	NO	14
MV-304	P-100B discharge	70.0	21.4	2.9	grade	NO	14
MV-305	DR	66.3	21.4	1.3	grade	NC	14
MV-306	benzene feed	16.4	30.9	1.0	grade	NO	8
MV-307	benzene feed	18.0	30.9	1.0	grade	NO	8
MV-308	benzene feed bypass	17.2	30.9	1.8	grade	NC	8
HV-116	start-up solvent charge	24.0	30.4	1.2	grade	NC	8
HV-117	catalyst charge	25.8	43.2	14.3	PF-R140	NC	5
HV-113	E-103 cooling water supply	20.0	33.6	9.0	PF-R80	NO	3
MV-309	E-103 cooling water return	20.0	34.4	19.5	PF-E185	LO	6
MV-310	DR	21.0	34.4	18.9	PF-E185	NC	6
MV-311	R-101 overflow to R-102	25.4	39.0	9.1	PF-R80	NO	3
MV-312	R-101 overflow to R-102	27.4	39.0	8.9	PF-R80	NO	3
MV-313	R-101 overflow to R-102 bypass	26.3	39.0	9.6	PF-R80	NC	3
MV-314	hydrogen to R-101 sparger	27.8	31.6	1.0	grade	NO	8
MV-315	hydrogen to R-101 sparger	30.6	31.6	1.0	grade	NO	8
MV-316	hydrogen to R-101 sparger bypass	29.2	31.6	1.8	grade	NC	8
MV-317	P-101A suction	21.0	30.2	0.9	grade	NO	9
MV-318	P-101A discharge	21.0	28.7	3.3	grade	NO	9
MV-319	P-101B suction	26.0	30.2	0.9	grade	NO	9
MV-320	P-101B discharge	26.0	28.7	3.3	grade	NO	9
MV-321	DR	22.5	31.0	3.2	grade	NC	9
MV-322	D-101 continuous blowdown	6.5	43.0	8.6	PF-D80	THR	2
HV-110	MP steam to E-101 shell (start-up heating)	13.4	33.6	1.2	grade	NC	1
HV-121	loop vent to flare	33.6	37.2	11.6	PF-R2T	NC	4
MV-401	P-102A suction	32.0	30.2	0.6	grade	NO	9
MV-402	P-102A discharge	32.0	28.6	2.4	grade	NO	9
MV-403	P-102B suction	35.5	30.2	0.6	grade	NO	9
MV-404	P-102B discharge	35.5	28.6	2.4	grade	NO	9
MV-405	DR	32.0	30.7	0.4	grade	NC	9
MV-406	F-102A inlet	31.0	18.0	1.1	grade	NO	11
MV-407	F-102A outlet	31.0	15.6	1.2	grade	NO	11
MV-408	F-102B inlet	33.5	18.0	1.1	grade	NC	11
MV-409	F-102B outlet	33.5	15.6	1.2	grade	NC	11
MV-410	R-102 liquid to R-103	30.2	14.6	1.0	grade	NO	11
MV-411	R-102 liquid to R-103	28.6	14.6	1.0	grade	NO	11
MV-412	R-102 liquid to R-103 bypass	29.4	14.6	1.7	grade	NC	11

Tag	Service	x	y	EL	From	Normal	Station
MV-413	E-106 cooling water supply	36.0	1.5	0.9	grade	NO	12
MV-414	E-106 cooling water return	36.0	2.6	3.2	grade	LO	12
MV-415	make-up hydrogen	54.6	17.2	1.0	grade	NO	13
MV-416	make-up hydrogen	55.6	17.2	1.0	grade	NO	13
MV-417	make-up hydrogen bypass	55.1	17.2	1.7	grade	NC	13
HV-120	nitrogen purge to the loop	55.2	16.4	1.0	grade	NC	13
MV-418	K-101 discharge	61.5	12.2	1.8	grade	NO	13
MV-419	product to storage	47.2	9.6	0.9	grade	NO	12
MV-420	product to storage	49.6	9.6	0.9	grade	NO	12
MV-421	product to storage bypass	48.4	9.6	1.6	grade	NC	12

6 · Isometrics — ISO-01 Benzene feed: P-100A/B → FV-101 / XV-101 → R-101 N1

ISO-01 · Benzene feed: P-100A/B → FV-101 / XV-101 → R-101 N1

Sketch, not to scale: straight runs opened or shortened (drawn 2.2·L^0.62); every elevation EL is true (m above grade); routes and positions from layout.json



Cyclohexane Reactor Ver 1 OTS Issued for teaching · numbers from the simulator model	**CHX-V1-PL-PP-001** · sheet ISO-01 Benzene feed: P-100A/B → FV-101 / XV-101 → R-101 N1	Scale not to scale Units m · EL above grade 0.0 · plant north = true north (assumed)	Rev 0 · 2026-10-27 Drawn Apinan Soottitantawat · checked — · approved —

ISO-01 · bill of materials (per line, from the route)

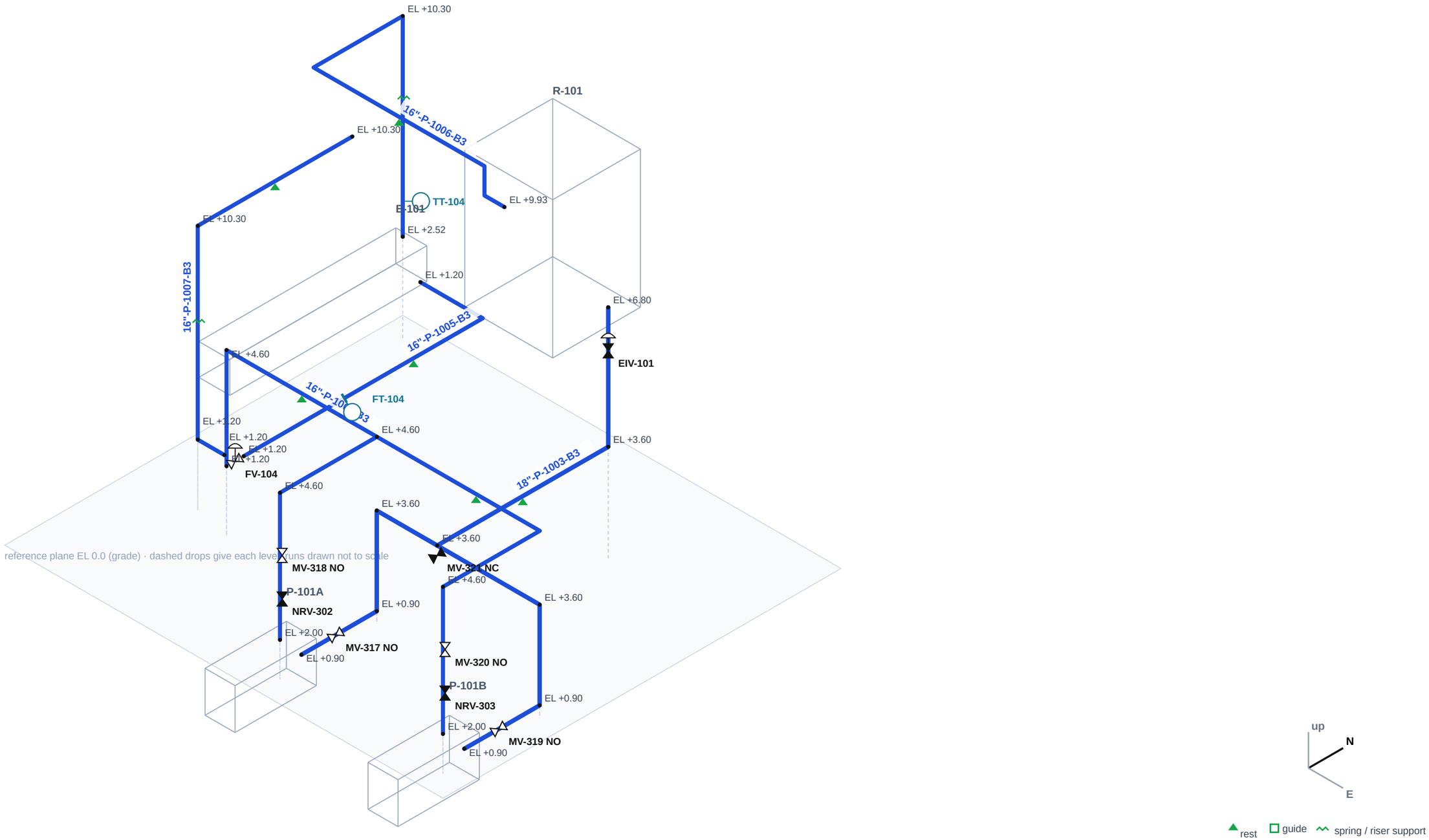
Line	P&ID	Spec / class / sch	Pipe m	Elbows	Tees	WN flanges	Gaskets / bolts	Valves and inline items	Branch items	Supports R/G/S
6"-P-1001-B1	6"-P-1001	B1 · 300 · 40	91.5	3	1	14	8	EIV-100, MV-301 (NO), MV-303 (NO)	–	14 / 5 / 0
3"-P-1002-B1	4"-P-1002	B1 · 300 · 40	104.4	14	2	42	22	NRV-100A, MV-302 (NO), NRV-100B, MV-304 (NO), MV-306 (NO), MV-307 (NO), MV-308 (NC), FV-101, XV-101, FT-101	–	19 / 9 / 0

Sketch quantities from the route (no spool breaks or field welds); supports at the typical span for the size (standard wall, water-filled), a guide every third support, a riser spring on long hot verticals (assumed).

6.2 · ISO-02 Pump-around loop: R-101 → P-101A/B → FV-104 → E-101 / by-pass → R-101

ISO-02 · Pump-around loop: R-101 → P-101A/B → FV-104 → E-101 / by-pass → R-101

Sketch, not to scale: straight runs opened or shortened (drawn 2.2·L^0.62); every elevation EL is true (m above grade); routes and positions from layout.json



Cyclohexane Reactor Ver 1 OTS Issued for teaching · numbers from the simulator model	**CHX-V1-PL-PP-001** · sheet ISO-02 Pump-around loop: R-101 → P-101A/B → FV-104 → E-101 / by-pass → R-101	Scale not to scale Units m · EL above grade 0.0 · plant north = true north (assumed)	Rev 0 · 2026-10-27 Drawn Apinan Soottitantawat · checked — · approved —

ISO-02 · bill of materials (per line, from the route)

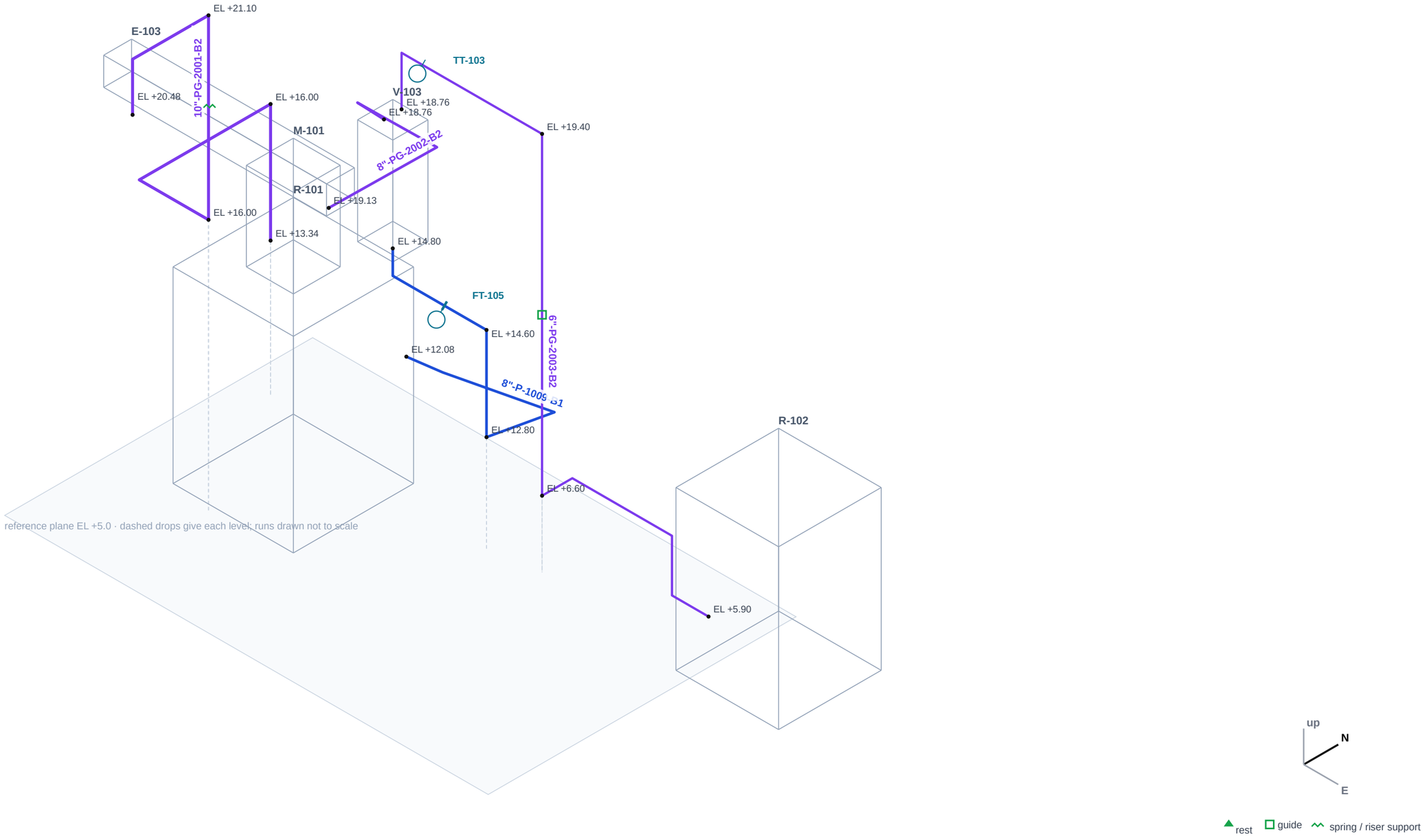
Line	P&ID	Spec / class / sch	Pipe m	Elbows	Tees	WN flanges	Gaskets / bolts	Valves and inline items	Branch items	Supports R/G/S
18"-P-1003-B3	18"-SL-1010	B3 · 300 · 40	24.4	5	2	14	8	EIV-101, MV-317 (NO), MV-319 (NO)	–	1 / 0 / 0
16"-P-1004-B3	16"-SL-1011	B3 · 300 · 40	26.5	5	1	18	10	NRV-302, MV-318 (NO), NRV-303, MV-320 (NO)	–	2 / 0 / 0
16"-P-1005-B3	12"-SL-1015	B3 · 300 · 40	10.7	1	0	6	4	FT-104	–	2 / 0 / 0
16"-P-1006-B3	16"-SL-1016	B3 · 300 · 40	19.2	4	0	2	2	–	TT-104	1 / 0 / 1
16"-P-1007-B3	16"-SL-1013	B3 · 300 · 40	16.3	2	0	2	2	–	–	1 / 0 / 1

Sketch quantities from the route (no spool breaks or field welds); supports at the typical span for the size (standard wall, water-filled), a guide every third support, a riser spring on long hot verticals (assumed).

6.3 · ISO-03 Overhead, reflux and vent: R-101 → E-103 → V-103 → reflux to R-101, gas to R-102

ISO-03 · Overhead, reflux and vent: R-101 → E-103 → V-103 → reflux to R-101, gas to R-102

Sketch, not to scale: straight runs opened or shortened (drawn 2.2·L^0.62); every elevation EL is true (m above grade); routes and positions from layout.json



Cyclohexane Reactor Ver 1 OTS Issued for teaching · numbers from the simulator model	**CHX-V1-PL-PP-001** · sheet ISO-03 Overhead, reflux and vent: R-101 → E-103 → V-103 → reflux to R-101, gas to R-102	Scale not to scale Units m · EL above grade 0.0 · plant north = true north (assumed)	Rev 0 · 2026-10-27 Drawn Apinan Soottitantawat · checked — · approved —

ISO-03 · bill of materials (per line, from the route)

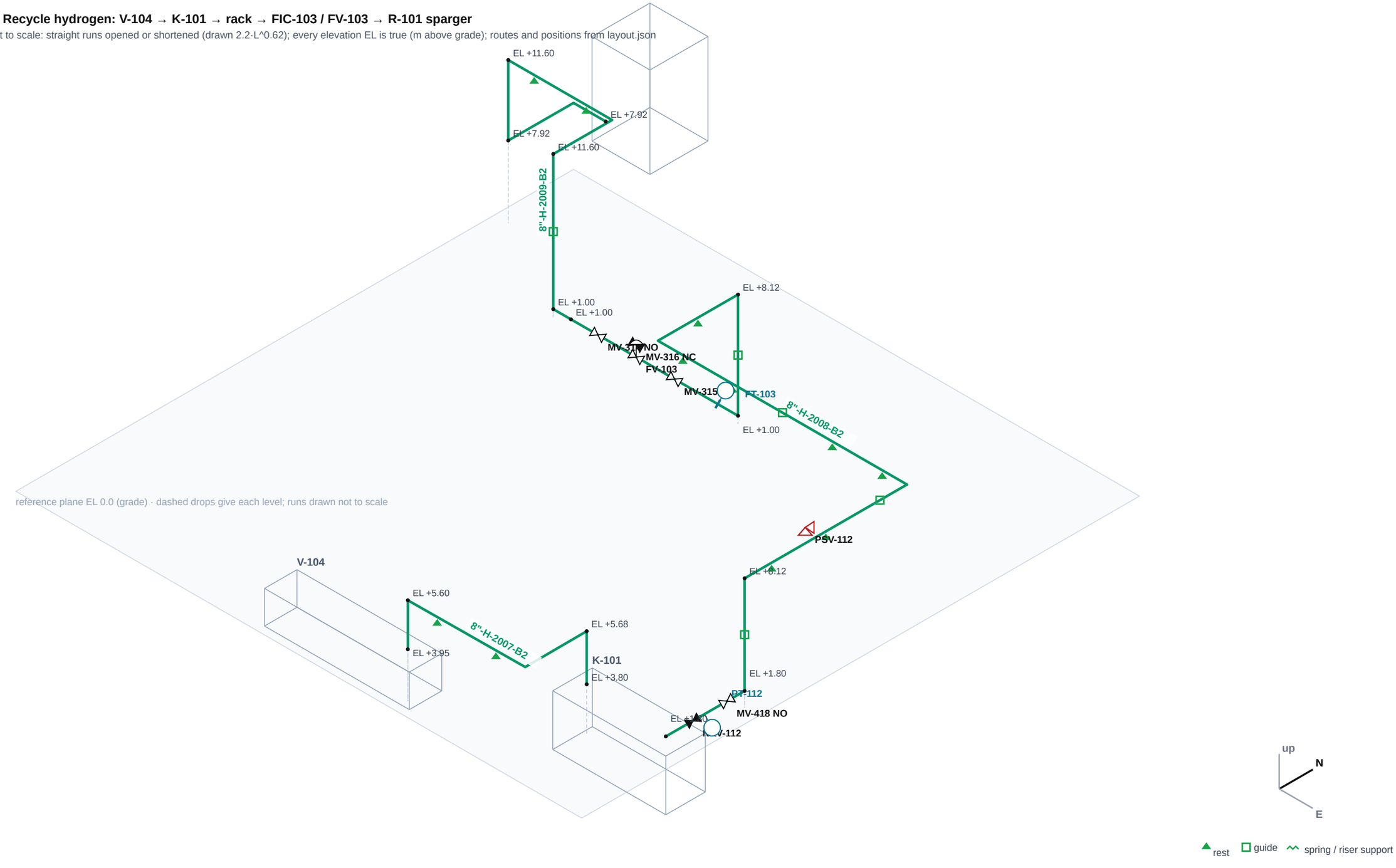
Line	P&ID	Spec / class / sch	Pipe m	Elbows	Tees	WN flanges	Gaskets / bolts	Valves and inline items	Branch items	Supports R/G/S
10"-PG-2001-B2	10"-P-1005	B2 · 300 · 40	13.9	5	0	2	2	–	–	0 / 0 / 1
8"-PG-2002-B2	6"-P-1006	B2 · 300 · 40	4.1	2	0	2	2	–	–	0 / 0 / 0
8"-P-1009-B1	6"-P-1007	B1 · 300 · 40	7.2	4	0	6	4	FT-105	–	0 / 0 / 0
6"-PG-2003-B2	6"-G-1008	B2 · 300 · 40	19.7	6	0	2	2	–	TT-103	0 / 1 / 0

Sketch quantities from the route (no spool breaks or field welds); supports at the typical span for the size (standard wall, water-filled), a guide every third support, a riser spring on long hot verticals (assumed).

6.4 · ISO-04 Recycle hydrogen: V-104 → K-101 → rack → FIC-103 / FV-103 → R-101 sparger

ISO-04 · Recycle hydrogen: V-104 → K-101 → rack → FIC-103 / FV-103 → R-101 sparger

Sketch, not to scale: straight runs opened or shortened (drawn 2.2·L^0.62); every elevation EL is true (m above grade); routes and positions from layout.json



Cyclohexane Reactor Ver 1 OTS Issued for teaching · numbers from the simulator model	**CHX-V1-PL-PP-001** · sheet ISO-04 Recycle hydrogen: V-104 → K-101 → rack → FIC-103 / FV-103 → R-101 sparger	Scale not to scale Units m · EL above grade 0.0 · plant north = true north (assumed)	Rev 0 · 2026-10-27 Drawn Apinan Soottitantawat · checked — · approved —

ISO-04 · bill of materials (per line, from the route)

Line	P&ID	Spec / class / sch	Pipe m	Elbows	Tees	WN flanges	Gaskets / bolts	Valves and inline items	Branch items	Supports R/G/S
8"-H-2007-B2	10"-G-2007	B2 · 300 · 40	15.1	3	0	2	2	–	–	2 / 0 / 0
8"-H-2008-B2	8"-G-2008	B2 · 300 · 40	68.9	6	0	26	14	MV-314 (NO), MV-315 (NO), FV-103, NRV-112, MV-418 (NO), FT-103	PT-112	8 / 4 / 0
8"-H-2009-B2	8"-G-1009	B2 · 300 · 40	28.8	6	0	2	2	–	–	2 / 1 / 0

Sketch quantities from the route (no spool breaks or field welds); supports at the typical span for the size (standard wall, water-filled), a guide every third support, a riser spring on long hot verticals (assumed).

6.5 · ISO-05 Relief and flare: PSV tailpipes, EDV-101 / HV-121 and the unit flare header to V-190

ISO-05 · Relief and flare: PSV tailpipes, EDV-101 / HV-121 and the unit flare header to V-190

Sketch, not to scale: straight runs opened or shortened (drawn 2.2·L^0.62); every elevation EL is true (m above grade); routes and positions from layout.json



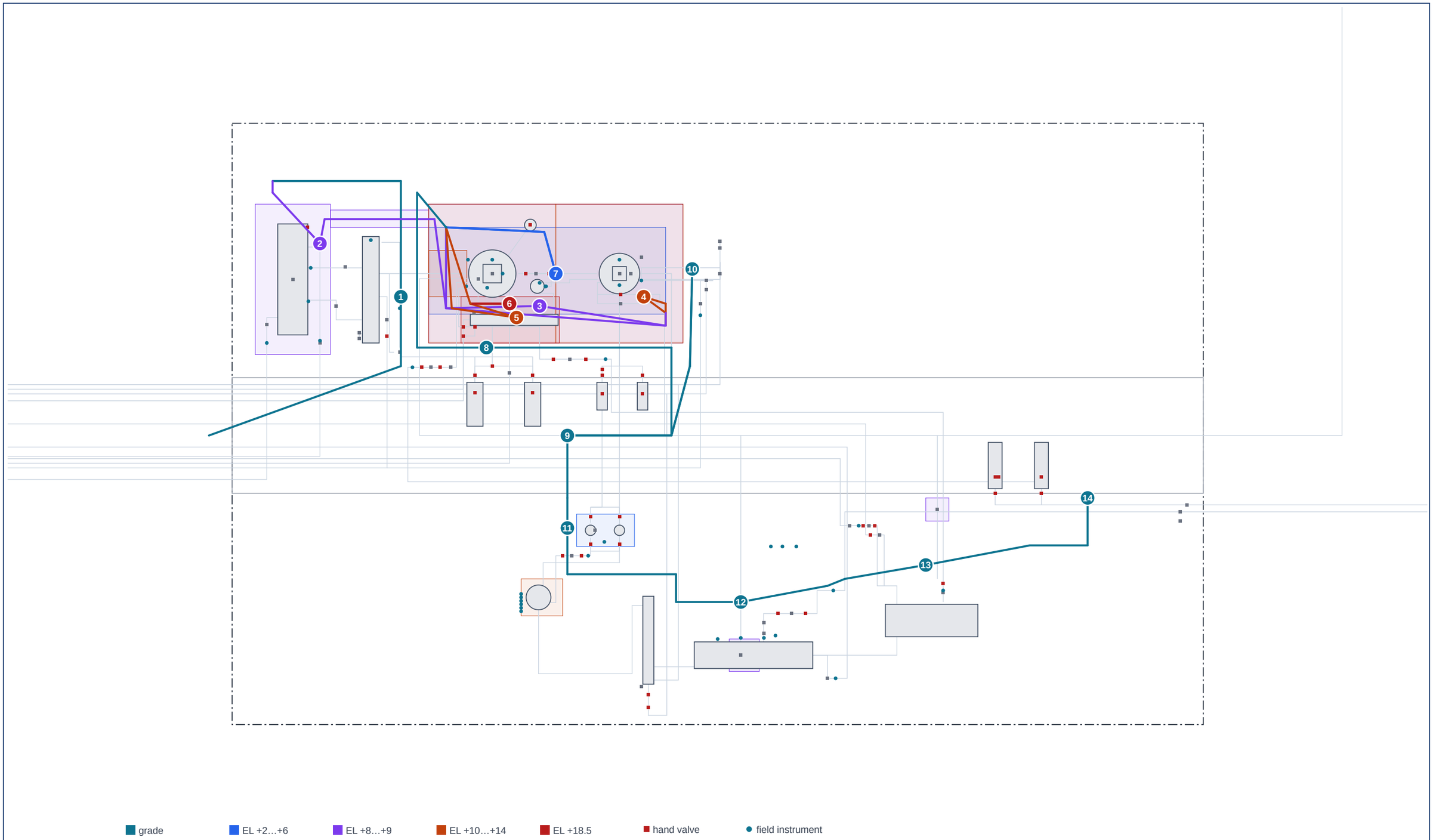
Cyclohexane Reactor Ver 1 OTS Issued for teaching · numbers from the simulator model	**CHX-V1-PL-PP-001** · sheet ISO-05 Relief and flare: PSV tailpipes, EDV-101 / HV-121 and the unit flare header to V-190	Scale not to scale Units m · EL above grade 0.0 · plant north = true north (assumed)	Rev 0 · 2026-10-27 Drawn Apinan Soottitantawat · checked — · approved —

ISO-05 · bill of materials (per line, from the route)

Line	P&ID	Spec / class / sch	Pipe m	Elbows	Tees	WN flanges	Gaskets / bolts	Valves and inline items	Branch items	Supports R/G/S
6"-FL-1040 RLB 24" (P&ID 6" HOLD)	6"-FL-1040	A2 · 150 · 40	20.5	3	0	6	4	PSV-101	–	2 / 0 / 0
8"-FL-2016 RLB 10" / 8" / 8" (P&ID 8" HOLD)	8"-FL-2016	A2 · 150 · 40	43.2	1	2	14	8	PSV-102, PSV-104, PSV-112	–	7 / 0 / 0
8"-FL-3013-A2	4"-FL-2015	A2 · 150 · 40	22.9	7	0	6	4	EDV-101	–	4 / 0 / 0
12"-FL-3014-A2	4"-FL-2015	A2 · 150 · 40	6.2	1	0	6	4	HV-121 (NC)	–	1 / 0 / 0
30"-FL-3015 (proposed)	30"-FL-3015 (proposed)	A2 · 150 · 40	176.0	3	0	2	2	–	–	14 / 6 / 0

Sketch quantities from the route (no spool breaks or field welds); supports at the typical span for the size (standard wall, water-filled), a guide every third support, a riser spring on long hot verticals (assumed).

7 · layout.json and the outside operator's round (sheet DWG-6)



grade · EL +2…+6 · EL +8…+9 · EL +10…+14 · EL +18.5 · hand valve · field instrument

Cyclohexane Reactor Ver 1 OTS Issued for teaching · numbers from the simulator model	**CHX-V1-PL-PP-001** · sheet DWG-6 layout.json top view with the outside operator's round	Scale not to a fixed scale Units m · EL above grade 0.0 · plant north = true north (assumed)	Rev 0 · 2026-10-27 Drawn Apinan Soottitantawat · checked — · approved —

7.1 · Route stations

#	Station	Level	Stand at x, y, floor EL	Look at	Read and check
1	Steam bay at grade: E-101 and its valves	grade	14.6, 37.0, 0.0	12.0, 37.0, 1.6	HV-110, HY-101D, FV-104, FT-104, TT-104, HY-101A, HY-101B, HY-101C, ST-101
2	D-101 steam-drum deck EL +8.0	PF-D80	7.6, 41.6, 8.0	5.2, 38.5, 9.6	LT-109, PT-110, FT-109, FT-111, LV-109, PV-110, MV-322, PSV-109
3	Reactor floor EL +8.0: R-101 and R-102 side nozzles, LV-101 station, HV-113	PF-R80	26.6, 36.2, 8.0	26.3, 39.0, 9.0	TT-100, TT-101, TT-102, HV-113, MV-311, LV-101, MV-312, MV-313, TSV-102J
4	R-102 top platform EL +10.8	PF-R2T	35.6, 37.0, 10.8	33.6, 37.0, 11.6	PT-102, LT-102, PSV-102, EDV-101, HV-121
5	Reactor top deck EL +14.0: agitator, PSV-101, catalyst pot, reflux drum	PF-R140	24.6, 35.2, 14.0	22.5, 39.0, 14.5	PT-101, LT-101, SSL-101, PSV-101, HV-117, LG-103, FT-105
6	Condenser deck EL +18.5: E-103	PF-E185	24.0, 36.4, 18.5	24.4, 35.0, 19.8	TT-103, MV-309, TSV-113, MV-310
7	Reactor bottoms EL +4.5: emergency isolation valves	PF-R45	28.0, 39.0, 4.5	22.5, 39.0, 6.1	EIV-101, EIV-102
8	Valve strip at grade: benzene feed station, start-up solvent, H_2 to the sparger	grade	22.0, 32.6, 0.0	20.0, 31.2, 1.0	FT-101, MV-306, FV-101, MV-307, MV-308, XV-101, HV-116, NRV-301, FT-103, MV-314, FV-103, MV-315, MV-316
9	Pump row in the rack aisle: P-101A/B and P-102A/B	grade	29.0, 25.0, 0.0	26.0, 28.0, 1.0	MV-317, NRV-302, MV-318, MV-319, NRV-303, MV-320, MV-321, MV-401, NRV-401, MV-402, MV-403, NRV-402, MV-404, MV-405
10	R-102 jacket station (east of the structure)	grade	39.8, 39.4, 0.0	41.4, 38.4, 1.2	FT-108, HY-102A, HY-102B, TV-102, HY-102C, HY-102D, ST-102
11	Catalyst filters, LV-102 and the R-103 base	grade	29.0, 17.0, 0.0	31.5, 16.8, 1.5	PDT-102, MV-406, MV-407, MV-408, MV-409, TSV-102F, FT-102, MV-410, LV-102, MV-411, MV-412, TT-106, TT-107A, TT-107B, TT-107C, TT-107D, TT-107E
12	E-106, V-104, product station and analyser shelter	grade	44.0, 10.6, 0.0	45.0, 7.0, 1.8	MV-413, MV-414, TSV-106, TT-109, LT-104, LT-104S, PT-105, EIV-104, XV-104, MV-419, LV-104, MV-420, MV-421, FT-110, HV-107, FT-107, PSV-104, AT-100, AT-103, AT-105
13	K-101 compressor shelter and the make-up hydrogen station	grade	60.0, 13.8, 0.0	60.5, 9.0, 1.8	PT-112, NRV-112, MV-418, PSV-112, FT-106, XV-105, MV-415, PV-105, MV-416, MV-417, HV-120, NRV-403
14	Feed pumps P-100A/B and the east battery limit	grade	74.0, 19.6, 0.0	68.0, 21.5, 1.0	MV-301, NRV-100A, MV-302, MV-303, NRV-100B, MV-304, MV-305, EIV-100, HY-115A, HY-115B

7.2 · layout.json keys

Key	Content	Count
equipment[]	tag, eql, name, type (vcyl · hcyl · exchanger · column · pump · compressor · box · tank), c3, box [x0,y0,z0,x1,y1,z1], D, TT, hd, zb, BTL, TTL, ztop, zc, L, axis, color, on, support, basis, nozzles[] {id, svc, ll, pid, size, p}	26
platforms[], stairs[]	platform {id, name, z, x0, y0, x1, y1, holes}; stair {x0…y1, levels}; ladder {x, y, levels}	13 / 13
pipes[]	line (LL), pid (P&ID), svc, nps, spec, from, to, segs (polylines of [x,y,z]), iso	52
valves[]	tag, type, hand, normal (NO · NC · LO · THR · FC · FO · auto · set), svc, sheet, p, from (platform or grade), model	98
instruments[]	tag, svc, type, loc, sheet, p, from, note	40
route[]	n, name, p (floor point), eye 1.6, look, via (waypoints), level, items	14
buildings, structures, rack, roads, areas, zones, calc, isos, spacing, eq, src, assumed	site objects, hazardous-area sources (release elevation, radii), hydraulics inputs, isometric definitions, rules, queries, copies of the source data	

8 · Checks against the other documents

92 rows, 89 ✓, 3 ✗; the same functions run in the Layout & isometrics tab and in tools/validate_layout.js. Each ✗ names its engineering query.

Every EQL item on the plot plan

Item	This document	Limit / source		Query
R-101	R-101 (22.5, 39.0)	R-101	✓	CHX-V1-ME-EQL-001
M-101	M-101 (22.5, 39.0)	M-101	✓	CHX-V1-ME-EQL-001
R-102	R-102 (33.5, 39.0)	R-102	✓	CHX-V1-ME-EQL-001
M-102	M-102 (33.5, 39.0)	M-102	✓	CHX-V1-ME-EQL-001
R-103	R-103 (26.5, 11.0)	R-103	✓	CHX-V1-ME-EQL-001
E-101	E-101 (12.0, 37.6)	E-101	✓	CHX-V1-ME-EQL-001
E-103	E-103 (24.4, 35.0)	E-103	✓	CHX-V1-ME-EQL-001
E-106	E-106 (36.0, 7.3)	E-106	✓	CHX-V1-ME-EQL-001
D-101	D-101 (5.3, 38.5)	D-101	✓	CHX-V1-ME-EQL-001
D-102	D-102 (25.8, 43.2)	D-102	✓	CHX-V1-ME-EQL-001
V-103	V-103 (26.4, 37.9)	V-103	✓	CHX-V1-ME-EQL-001
V-104	V-104 (45.1, 6.0)	V-104	✓	CHX-V1-ME-EQL-001
F-102A/B	F-102A (31.0, 16.8), F-102B (33.5, 16.8)	F-102A, F-102B	✓	CHX-V1-ME-EQL-001
P-100A/B	P-100A (66.0, 22.4), P-100B (70.0, 22.4)	P-100A, P-100B	✓	CHX-V1-ME-EQL-001
P-101A/B	P-101A (21.0, 27.7), P-101B (26.0, 27.7)	P-101A, P-101B	✓	CHX-V1-ME-EQL-001
P-102A/B	P-102A (32.0, 28.4), P-102B (35.5, 28.4)	P-102A, P-102B	✓	CHX-V1-ME-EQL-001
K-101	K-101 (60.5, 9.0)	K-101	✓	CHX-V1-ME-EQL-001

Every line number on an isometric is in the line list or the EPC P&ID

Item	This document	Limit / source		Query
ISO-01 · 6"-P-1001-B1	LL ✓ · P&ID ✓	LL or P&ID	✓	
ISO-01 · 3"-P-1002-B1	LL ✓ · P&ID ✓	LL or P&ID	✓	
ISO-02 · 18"-P-1003-B3	LL ✓ · P&ID ✓	LL or P&ID	✓	
ISO-02 · 16"-P-1004-B3	LL ✓ · P&ID ✓	LL or P&ID	✓	
ISO-02 · 16"-P-1005-B3	LL ✓ · P&ID ✓	LL or P&ID	✓	
ISO-02 · 16"-P-1006-B3	LL ✓ · P&ID ✓	LL or P&ID	✓	
ISO-02 · 16"-P-1007-B3	LL ✓ · P&ID ✓	LL or P&ID	✓	
ISO-03 · 10"-PG-2001-B2	LL ✓ · P&ID ✓	LL or P&ID	✓	
ISO-03 · 8"-PG-2002-B2	LL ✓ · P&ID ✓	LL or P&ID	✓	
ISO-03 · 8"-P-1009-B1	LL ✓ · P&ID ✓	LL or P&ID	✓	
ISO-03 · 6"-PG-2003-B2	LL ✓ · P&ID ✓	LL or P&ID	✓	
ISO-04 · 8"-H-2007-B2	LL ✓ · P&ID ✓	LL or P&ID	✓	
ISO-04 · 8"-H-2008-B2	LL ✓ · P&ID ✓	LL or P&ID	✓	
ISO-04 · 8"-H-2009-B2	LL ✓ · P&ID ✓	LL or P&ID	✓	
ISO-05 · 6"-FL-1040	LL – · P&ID ✓	LL or P&ID	✓	EQ-ej-3
ISO-05 · 8"-FL-2016	LL – · P&ID ✓	LL or P&ID	✓	EQ-ej-3
ISO-05 · 8"-FL-3013-A2	LL ✓ · P&ID ✓	LL or P&ID	✓	
ISO-05 · 12"-FL-3014-A2	LL ✓ · P&ID ✓	LL or P&ID	✓	
ISO-05 · 30"-FL-3015 (proposed)	LL – · P&ID ✗	proposed number	✗	EQ-la-4

Spacing rules (typical practice, assumed)

Item	This document	Limit / source		Query
S1 Occupied control room (not blast-rated) to process equipment	85.9 m · D-101	≥ 60 m	✓	EQ-la-9
S2 Control room to atmospheric storage tanks	211.3 m · TK-106	≥ 60 m	✓	
S3 Substation / LER to process equipment (outside every zone)	34.0 m · D-101	≥ 30 m	✓	
S4 Process equipment to atmospheric flammable storage tanks	71.8 m · P-100B – TK-100	≥ 60 m	✓	
S5 Tank to tank, shell to shell (fixed roof, ≥ ½ the larger diameter)	4.0 m · TK-100 – TK-106: 13.0 m vs 9.0 m	≥ D/2	✓	
S6 Flare stack to process equipment and tanks (sterile radius, radiation study to confirm)	73.7 m · TK-100	≥ 60 m	✓	
S7 Hydrogen compressor to the reactors (large hot hydrocarbon inventories)	28.9 m · K-101 – R-103	≥ 15 m	✓	
S8 V-104 close to K-101 (short suction line, LSHH-104 protection)	6.3 m · V-104 – K-101	≤ 15 m	✓	
S9 Pumps handling flammables to the vessels they serve	7.4 m · P-101A – R-101	≥ 3 m	✓	
S10 Clear access between adjacent items (maintenance)	1.6 m · F-102A – F-102B	≥ 1.5 m	✓	
S11 Process equipment to the battery limit (roads outside)	3.5 m · E-106	≥ 2.5 m	✓	
S12 Tube bundle pulling / rodding space in front of each channel (tube length + 1 m; roads count)	21.7 m · E-103 30.0 ≥ 7.1 m · E-101 30.0 ≥ 8.3 m · E-106 30.0 ≥ 7.1 m	≥ L+1	✓	
S13 Two escape routes from every platform; travel to a stair or ladder ≤ 25 m	13.0 m · 11 walkable areas: 5 with ≥ 2 escapes, 6 small with one ladder	≤ 25 m	✓	
S14 Control room upwind: within ±45° of the prevailing-wind source (SW monsoon, from 225°)	25.4° · CR-101 bearing 250° from the unit centre; SW source 225°	≤ 45°	✓	EQ-la-9
S15 Atmospheric vent PSV-109: ≥ 3 m above any platform within 15 m	3.0 m · vent top EL +21.5; highest platform within 15 m EL +18.5 (PF-D80, PF-D112, PF-BR, PF-R45, PF-R80, PF-R110, PF-R140, PF-E185)	≥ 3 m	✓	
S16 Every PSV reachable from a platform or grade (maintenance)	5 PSVs · PSV-101 (PF-R140), PSV-109 (PF-D112), PSV-102 (PF-R2T), PSV-104 (PF-V104), PSV-112 (PF-K112)	every PSV	✓	
S17 Clear height under the pipe rack lower tier (vehicle and crane passage)	5.5 m · lower tier EL +6 less 0.5 m beam	≥ 5 m	✓	
S18 Analyser shelter within 20 m of its sample points (sample transport lag)	1.5 m · AS-101 to 10"-G-2007 / 6"-P-2011	≤ 20 m	✓	

Hazardous-area zones (CHX-V1-EE-HAC-001) on the plan

Item	This document	Limit / source		Query
CR-101 outside every zone (incl. the ground layer of heavy vapour)	clear by 81.6 m (nearest SR-03)	> 0	✓	
SS-101 outside every zone (incl. the ground layer of heavy vapour)	clear by 32.3 m (nearest SR-09)	> 0	✓	
No motor inside a Zone 1 envelope (3-D distance to the release point; motors are Ex ec / Ex pzc for Zone 2)	9 motors clear	none	✓	
Field instruments inside a zone (must be Ex ia IIC T4, as the schedule specifies)	33 of 40	info	✓	

layout.json validation (also tools/validate_layout.js)

Item	This document	Limit / source		Query
Every equipment-list item placed (A/B pairs as two items)	17 EQL tags → 21 placed items		✓	
No overlapping footprints at grade (items standing on grade)	17 grade items, pairwise clear		✓	
No intersecting equipment volumes (3-D boxes incl. insulation; a drive on its vessel excepted)	26 items		✓	
Equipment clear of buildings, stairs and ladders	clear		✓	
Every valve of the EPC P&ID and every field instrument has a position	98 valves (49 hand valves), 40 field instruments		✓	
Every hand valve operable from grade or a platform (≤ 1.6 m off the edge, −0.6…+2.6 m above the floor)	49 of 49		✓	
Every field instrument readable from grade, a platform or its building	40 of 40		✓	
Every platform reachable from grade by stairs or ladders	13 platforms		✓	
Route stations, waypoints and the walk between them at grade (clear of equipment) or on reachable platforms	14 stations, 43 waypoints		✓	
Station items exist and lie within 16 m (plan) of their station	137 items		✓	
Every hand valve and field instrument visited by the round (instruments in a building excepted)	137 of 138 on the route (in a building: JT-101)		✓	
Pipe routes clear of equipment bodies (1.2 m at each end excepted)	52 pipes, 2 384 m		✓	

Elevations from hydraulics

Item	This document	Limit / source		Query
P-101A/B NPSH available from R-101	5.57 m (bubble point, static 6.10 − losses 0.53 m)	≥ 5.20 m (NPSHr 4 m assumed)	✓	
P-102A/B NPSH available from R-102	3.91 m (bubble point, static 4.40 − losses 0.49 m)	≥ 3.00 m (NPSHr 2 m assumed)	✓	
P-100A/B NPSH available from TK-100	7.99 m (atmospheric tank − losses 0.37 m)	≥ 4.55 m (NPSHr 3.5 m assumed)	✓	
V-103 gravity reflux to R-101 N4	3.47 m of liquid above N4	≥ 2.75 m (ΔP 0.154 bar = 2.14 m + losses 0.12 m + margin 0.5 m)	✓	
R-101 → R-102 overflow (LV-101) self-draining	fall 0.85 m over 6.8 m, steepest-flat 10.2 %	≥ 1 % everywhere	✓	

Cross-document

Item	This document	Limit / source		Query
P-101 NPSH elevation: line list vs plot plan	LL 4 m static → 3.9 m; layout 6.10 m → 5.57 m	same basis	✗	EQ-la-1
Relief basis elevation of R-101 (fire case)	RLB 7.95 m · layout 7.95 m	± 0.5 m	✓	EQ-la-2

Item	This document	Limit / source		Query
Relief basis elevation of R-102 (fire case)	RLB 5.8 m · layout 5.80 m	± 0.5 m	✓	EQ-Ia-2
Relief basis elevation of R-103 (fire case)	RLB 2.49 m · layout 2.49 m	± 0.5 m	✓	EQ-Ia-2
Relief basis elevation of V-103 (fire case)	RLB 15.3 m · layout 15.00 m	± 0.5 m	✓	EQ-Ia-2
Relief basis elevation of V-104 (fire case)	RLB 1.45 m · layout 1.45 m	± 0.5 m	✓	EQ-Ia-2
Relief basis elevation of D-101 (fire case)	RLB 8.4 m · layout 8.30 m	± 0.5 m	✓	EQ-Ia-2
PSV-101 tailpipe length	layout 20.5 m	RLB 20.5 m (assumed)	✗	EQ-Ia-3
PSV-102 tailpipe length	layout 17.8 m	RLB 20.5 m (assumed)	✓	EQ-Ia-3
PSV-104 tailpipe length	layout 19.0 m	RLB 20.5 m (assumed)	✓	EQ-Ia-3
PSV-112 tailpipe length	layout 6.4 m	RLB 20.5 m (assumed)	✓	EQ-Ia-3
PSV-104 raised above the header: inlet 5.3 m	inlet loss ≈ 1.50 % of set (scaled from the RLB)	≤ 3 % (typical)	✓	EQ-Ia-3
PSV-112 raised above the header: inlet 6.1 m	inlet loss ≈ 1.64 % of set (scaled from the RLB)	≤ 3 % (typical)	✓	EQ-Ia-3
Unit flare header to V-190	layout 176 m	RLB 176 m	✓	EQ-Ia-3
PSV-109 vent stack	top EL +21.5 (≈ 9.9 m above the valve)	RLB tail 9.9 m (assumed)	✓	EQ-Ia-8
Valves of the EPC P&ID placed (both sheets)	98 placed	98 on the P&ID	✓	
Field instruments of the EPC P&ID placed	40 placed	40 on the P&ID	✓	

9 · Engineering queries and assumptions

ID	Query	Documents	Proposed resolution
EQ-la-1	P-101 NPSH: the line list assumed 4.0 m of static head Process engineering computed NPSHa ≈ 3.9 m with R-101 bottom only 4.0 m above the P-101 centre line (EP_AS.hNPSH, assumed). The plot plan puts the R-101 bottom at EL +7.0 and the P-101 centre line at EL +0.9: 6.1 m of static head, NPSHa as part 2 shows (bubble-point liquid, no vapour-pressure credit).	CHX-V1-PE-LL-001 (part 6 hydraulics), CHX-V1-ME-DS-P-101, this plot plan	LL rev 1 adopts the layout elevations; the P-101 datasheet states NPSHr ≤ 4.0 m at the rated 998 m³/h (assumed here) or the reactor is raised further.
EQ-la-2	Relief basis uses assumed elevations The relief load basis took bottom tangents R-101 4.0, R-102 2.0, R-103 1.5, V-104 1.0, V-103 9.0, D-101 8.0 m (assumed) for the fire case (wetted area up to 7.6 m above the fire level). The plot plan gives R-101 7.95, R-102 5.8, R-103 2.49, V-104 CL 2.6, V-103 15.3, D-101 CL 9.6 m; R-101 and R-102 stand on a structure with a solid floor at EL +8.0 only if the floor is plated (here: grating, so the fire level stays at grade).	CHX-V1-PE-RLB-001 (AS.geom), this plot plan, EQ-er-10	Relief & vent re-runs rlb_data.js with layout.json elevations (closes EQ-er-10); the R-101 and R-102 fire loads fall because less shell lies within 7.6 m of grade.
EQ-la-3	PSV tailpipe and header lengths differ from the relief basis The RLB assumed 20 m tailpipes, 1.5 m inlets and a 150 m unit header to V-190. On the plot plan the tailpipes, inlets and header have the lengths of part 3 (ISO-05). PSV-104 and PSV-112 are raised above the header so their tailpipes drain to it, which lengthens their inlet lines; the header is longer than 150 m because V-190 and the flare sit outside the tank-farm spacing.	CHX-V1-PE-RLB-001 (AS.tail, AS.inlet, AS.hdr), this plot plan	Relief & vent re-checks inlet loss (3 % rule) and built-up back pressure with the lengths of part 3; a 30" header number is proposed for LL rev 1 (EQ-er-9).
EQ-la-4	Flare header and PSV tailpipes have no line-list numbers The isometric of the relief system uses P&ID numbers 6"-FL-1040 and 8"-FL-2016 (both HOLD sizes: the RLB gives 24", 10", 8", 8") and the line-list numbers of the EDV-101 / HV-121 lines; the 30" unit header has no number anywhere.	CHX-V1-PE-LL-001, CHX-V1-PID-001 sheets 1–2, CHX-V1-PE-RLB-001	LL rev 1 adds the tailpipes and 30"-FL-3015-A2 (proposed here); P&ID rev 1 releases the HOLD sizes.
EQ-la-5	Make-up hydrogen tie-in point The plot plan follows the EPC P&ID (make-up into the K-101 suction, station beside the compressor shelter). The line list (6"-H-2010-B1 → V-104) and the PFD tie it into V-104 (EQ-px-3).	CHX-V1-PID-001-2, CHX-V1-PE-LL-001, CHX-V1-PFD-001	Process decides; LL rev 1 or P&ID rev 1 follows; the layout station does not move.
EQ-la-6	K-101 suction line cannot both drain to V-104 and enter a top nozzle The line list asks 10"/8"-H-2007 to slope back to V-104 (no liquid to K-101). With K-101 on a block foundation the suction nozzle (top, EL +3.8) is below the line run (EL +5.6), so the last 1.9 m drops into the machine.	CHX-V1-PE-LL-001, CHX-V1-ME-DS-K-101, this plot plan	K-101 vendor to confirm nozzle orientation; a suction strainer with a low-point drain upstream of the drop, or a table-top mounting with down-facing nozzles.
EQ-la-7	Tank sizes are not issued TK-100, TK-106 and TK-107 are outside the battery limit and have no datasheet; the plot plan sizes them for 48 h of storage (assumed) to check spacing and the bund.	this plot plan, CHX-V1-PM-DOC-001 (not issued)	Offsites issues tank datasheets; the bund height is re-checked (110 % of the largest tank).
EQ-la-8	PSV-109 vent height set by the platforms nearby PSV-109 discharges steam to atmosphere. The RLB assumed a 4 m tailpipe; the layout needs the vent outlet 3 m above the highest platform within 15 m (typical practice), so the vent stack is taller.	CHX-V1-PE-RLB-001 (AS.tail109), this plot plan	Relief & vent re-checks PSV-109 built-up back pressure with the vent length of part 2.
EQ-la-9	Control-room siting relies on distance only CR-101 is not blast-rated; it sits more than 60 m from the process and upwind of the SW monsoon (typical practice). During the NE monsoon it is downwind of the unit.	this plot plan, HAZID (H19 congestion review, H28), CHX-V1-EE-HAC-001	A facility-siting study (explosion overpressure at CR-101), positive-pressure HVAC and gas detection at the air intake.

Assumptions

- Every placement, elevation, route, platform and stair is this document's design (typical practice), not a vendor or site drawing.
- Spacing values are typical practice for teaching (insurance-type tables and siting practice), not a code requirement for a real site.
- Prevailing wind SW (May–October) and NE (November–February) for central Thailand, typical.
- Tank sizes for 48 h of storage; flare stack 45 m with a 60 m sterile radius; V-190 3.0 × 10 m.
- NPSHr of P-100, P-101, P-102 assumed (3.5, 4.0, 2.0 m) until the pump datasheets confirm them.
- E-103 shell pressure drop 0.10 bar (condensing, typical); vortex meter loss 2 velocity heads; reflux margin 0.5 m.
- Equivalent lengths of fittings: 90° long-radius elbow 16 D, tee branch 60 D, gate valve 8 D, ball/EIV 3 D, entrance 0.5 velocity head (typical).
- Pipe support spans for standard-wall pipe full of water (typical tables).
- Instrument transmitters remote-mounted at grade where the element sits high (R-103 multipoint, V-104 pressure), typical.
- Zone extents from the hazardous-area schedule CHX-V1-EE-HAC-001 drawn as circles round each source (plan view).